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(54) **CONSTANT STRAIN PEX EXPANSION TOOL HEAD**

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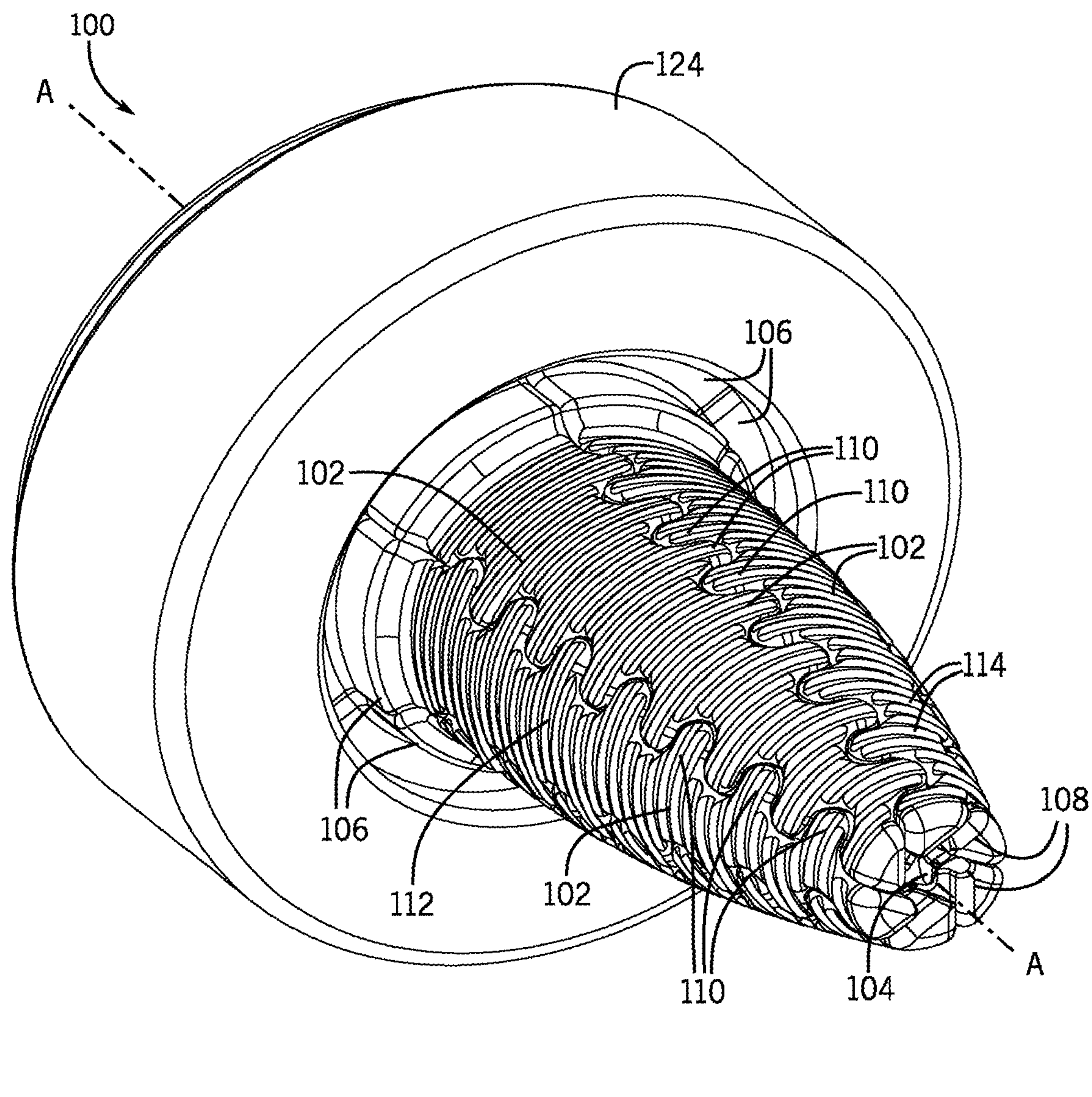
(52) **U.S. Cl.**

CPC ***B29C 57/04*** (2013.01); ***B29D 23/003*** (2013.01); *B29K 2023/0691* (2013.01); *B29L 2023/22* (2013.01)

(57) **ABSTRACT**

Various improved designs for expansion tool heads are disclosed herein that are structured (1) to provide improved and more-evenly distributed strain in the pipe during the expansion strokes of the tool head which can include more gradually inducing the strain in the pipe and/or (2) to avoid the induction of plastic deformation of the pipe in regions of seal formation between the radial-inwardly facing surface of the pipe and the radially-outward facing surface of the fittings (i.e., the ribs of the fitting) especially in instances in which the tool head is not rotated between separate expansion steps. This may be achieved, for example, by the use of convexly curved outer profiles that limit initial insertion depth of the tool in the pipe prior to expansions and/or by the use of a "hybrid" toothed profile.



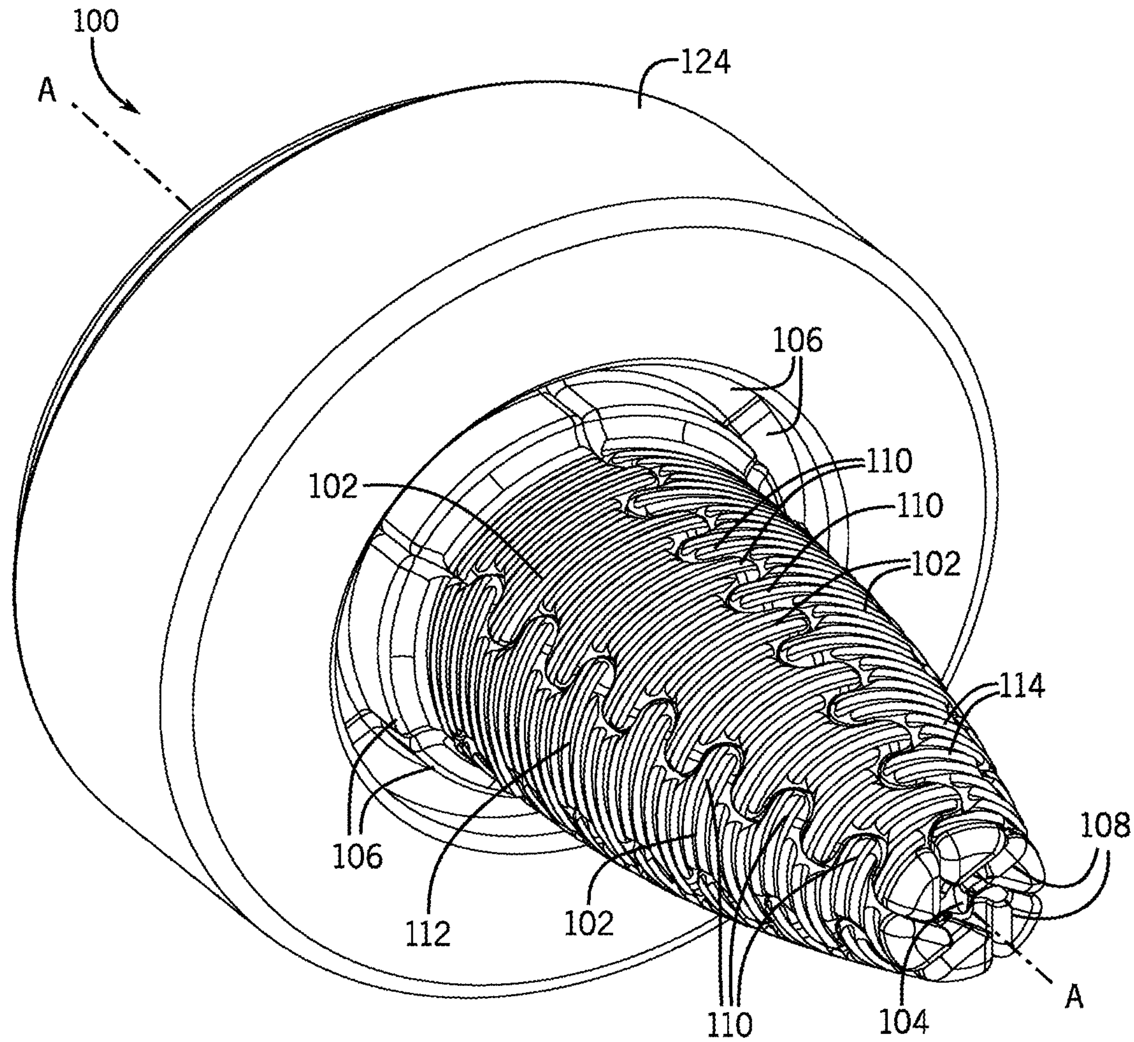


FIG. 1

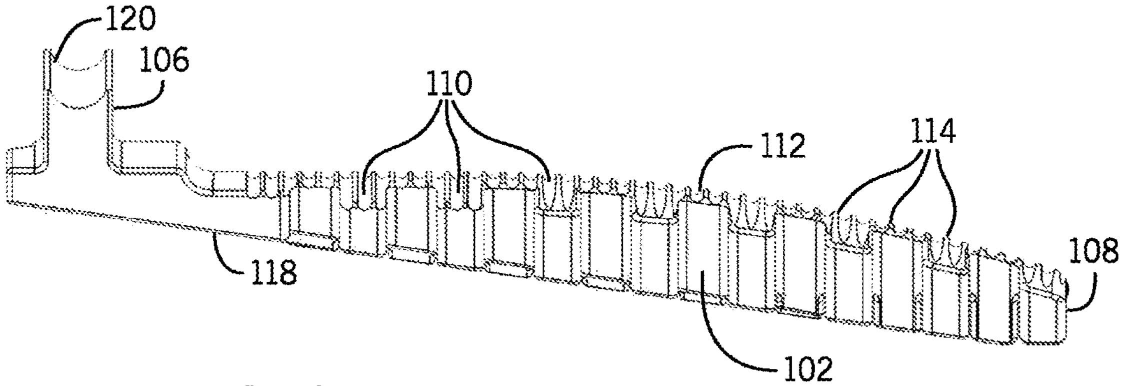


FIG. 2A

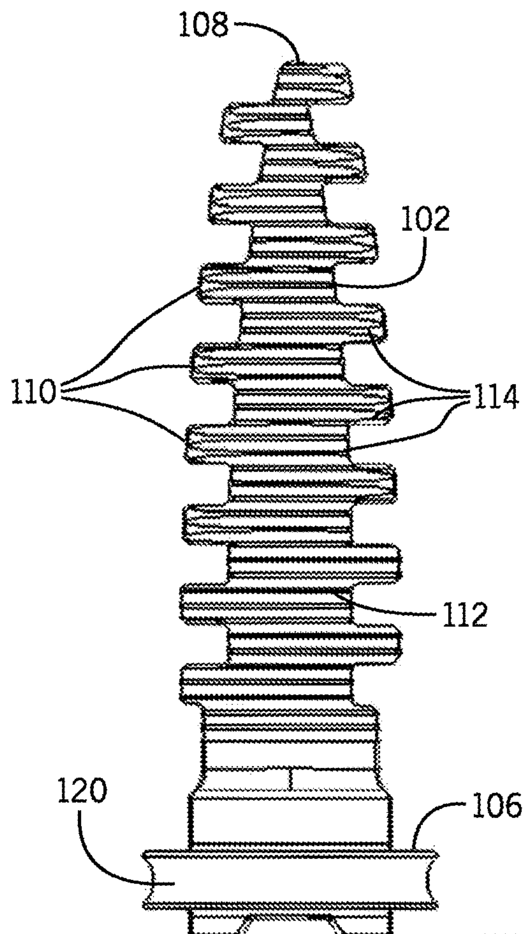


FIG. 2B

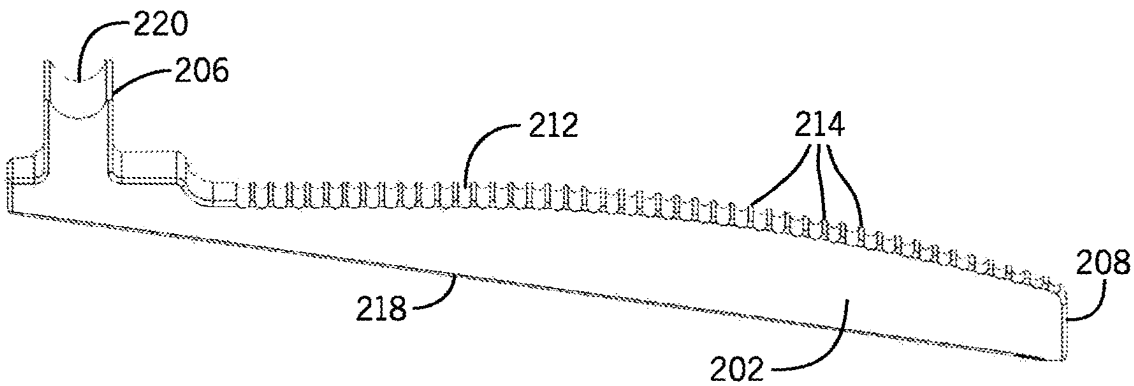


FIG. 3A

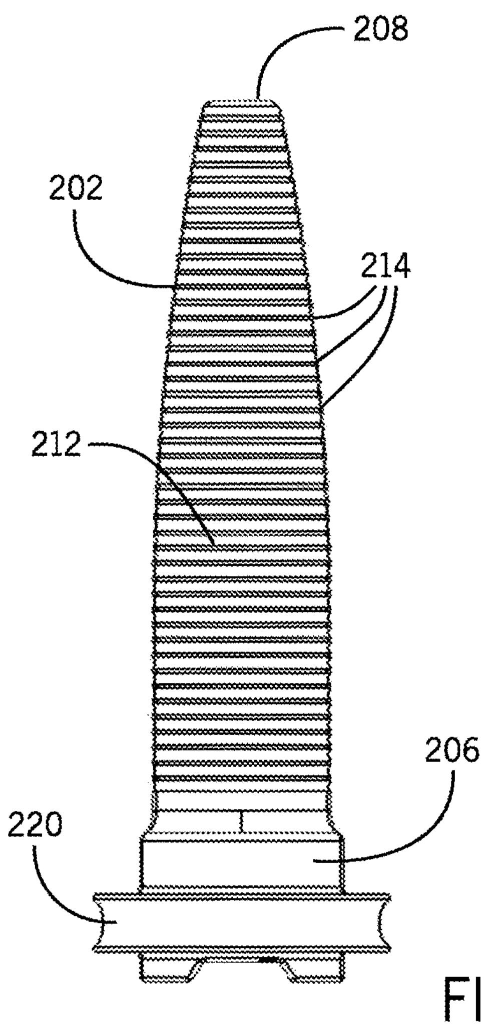


FIG. 3B

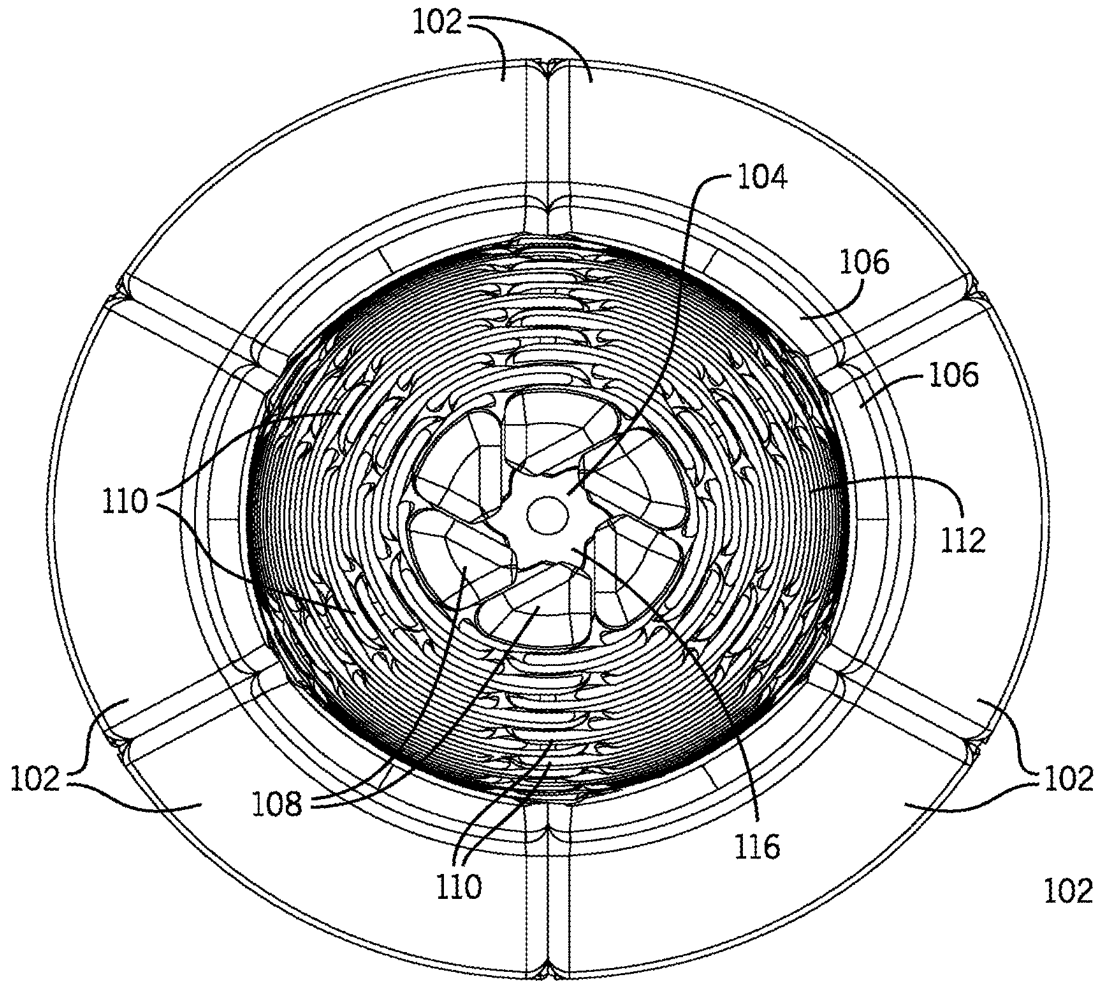


FIG. 4

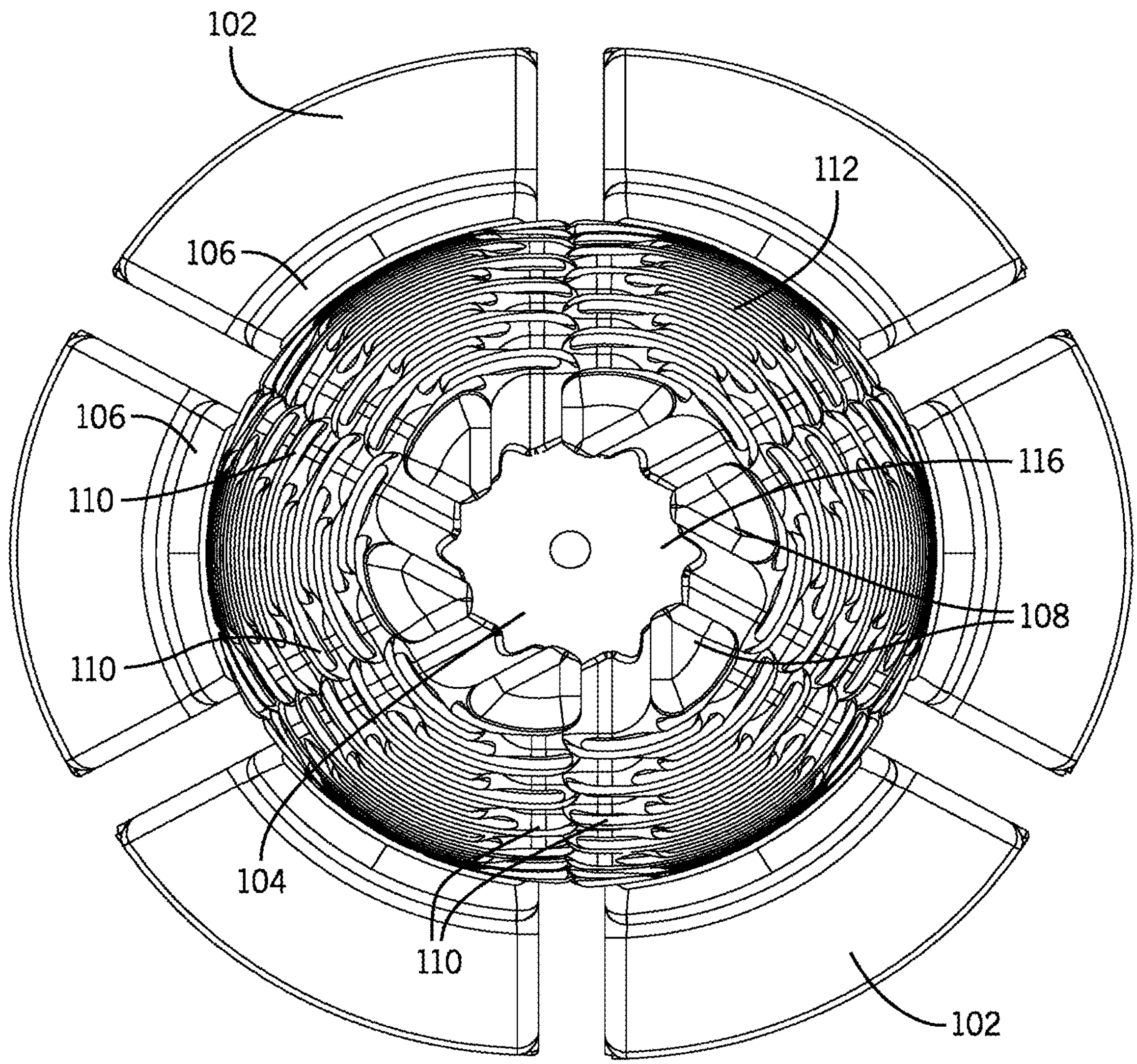


FIG. 5

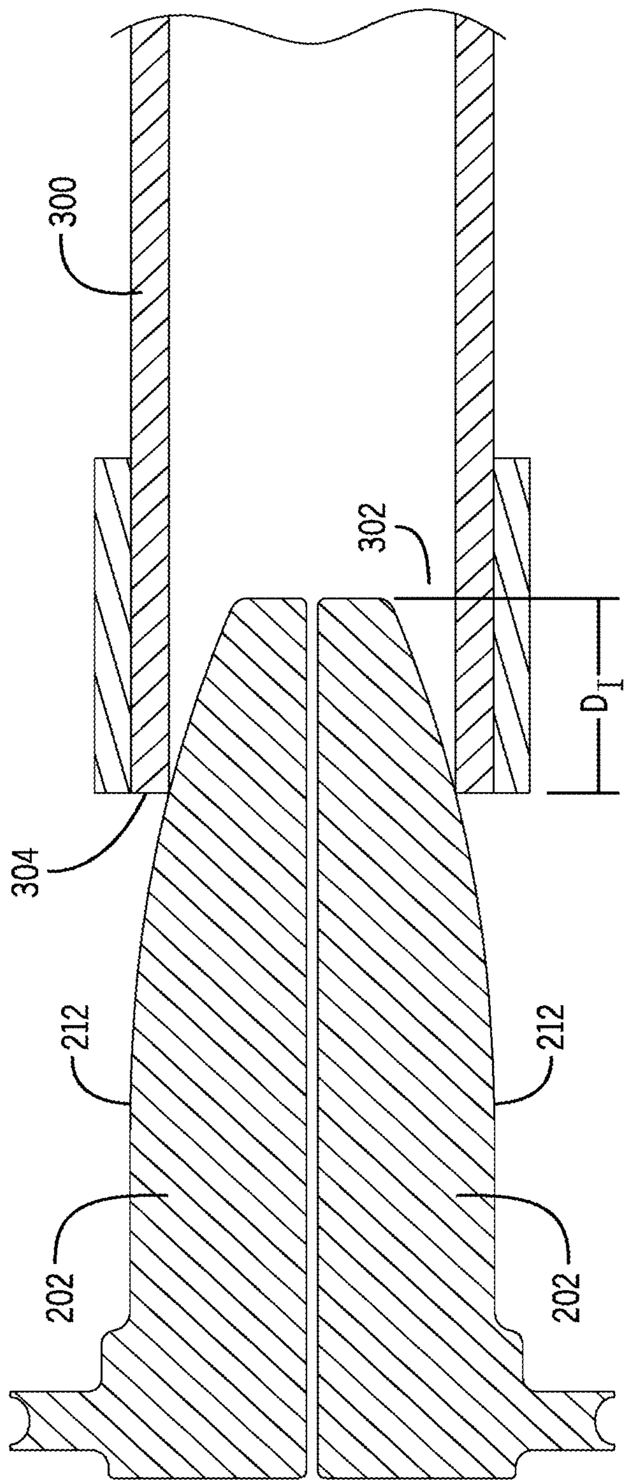


FIG. 6

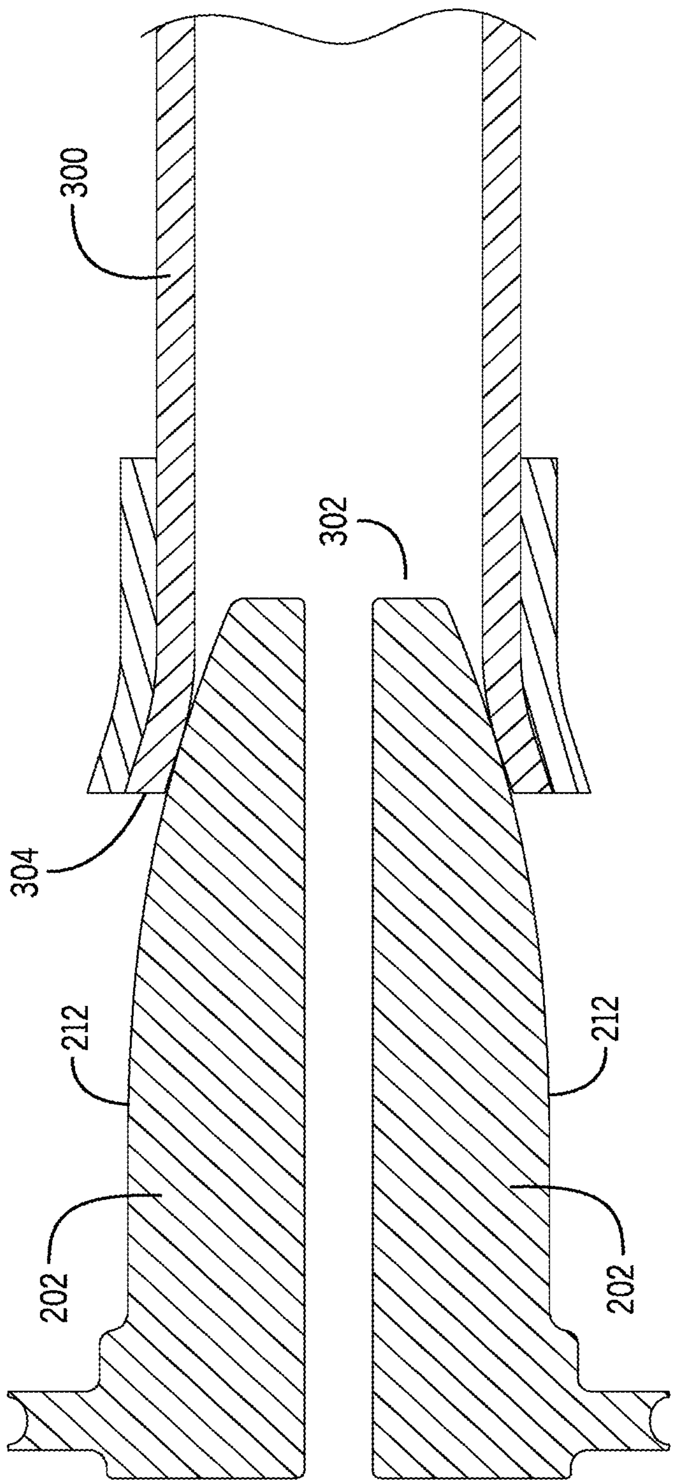


FIG. 7A

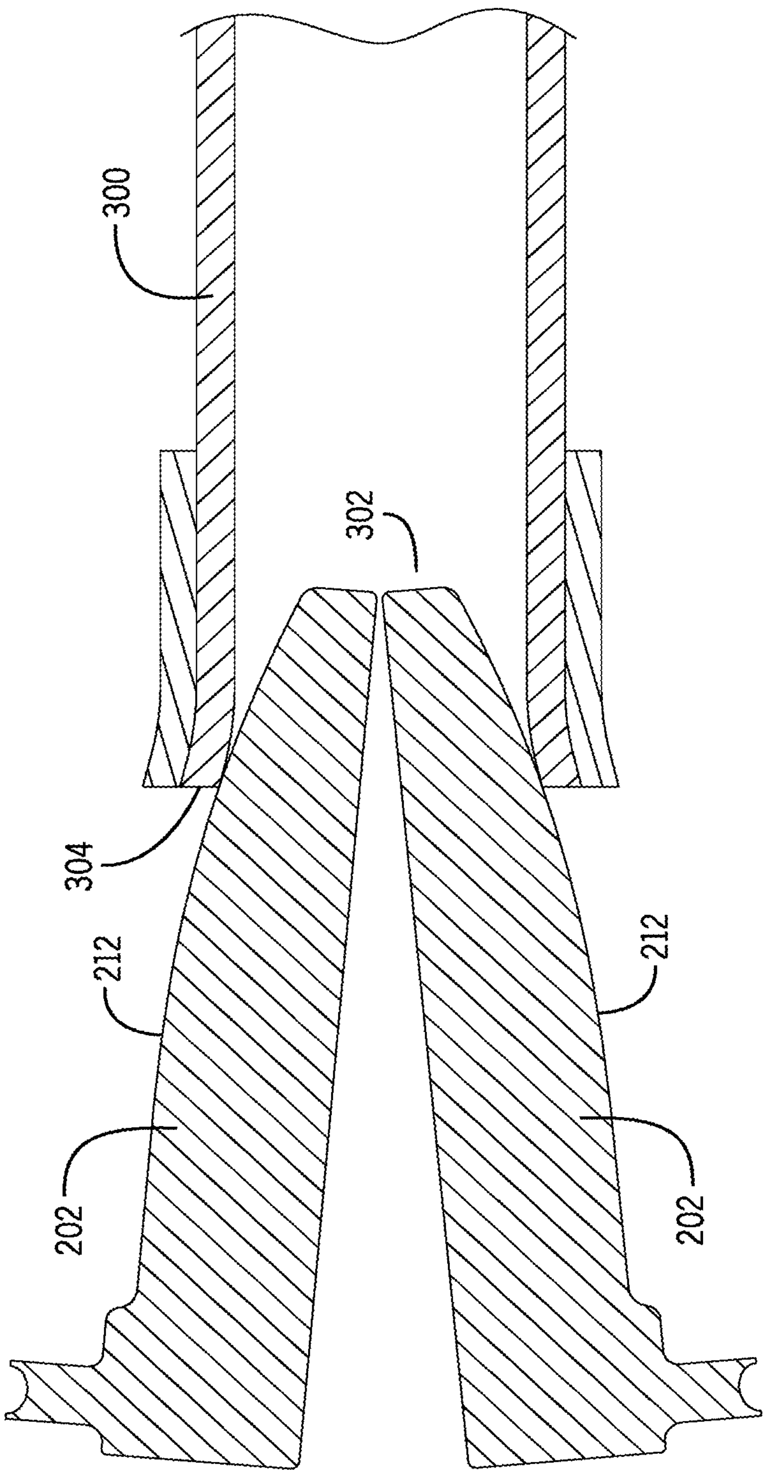


FIG. 7B

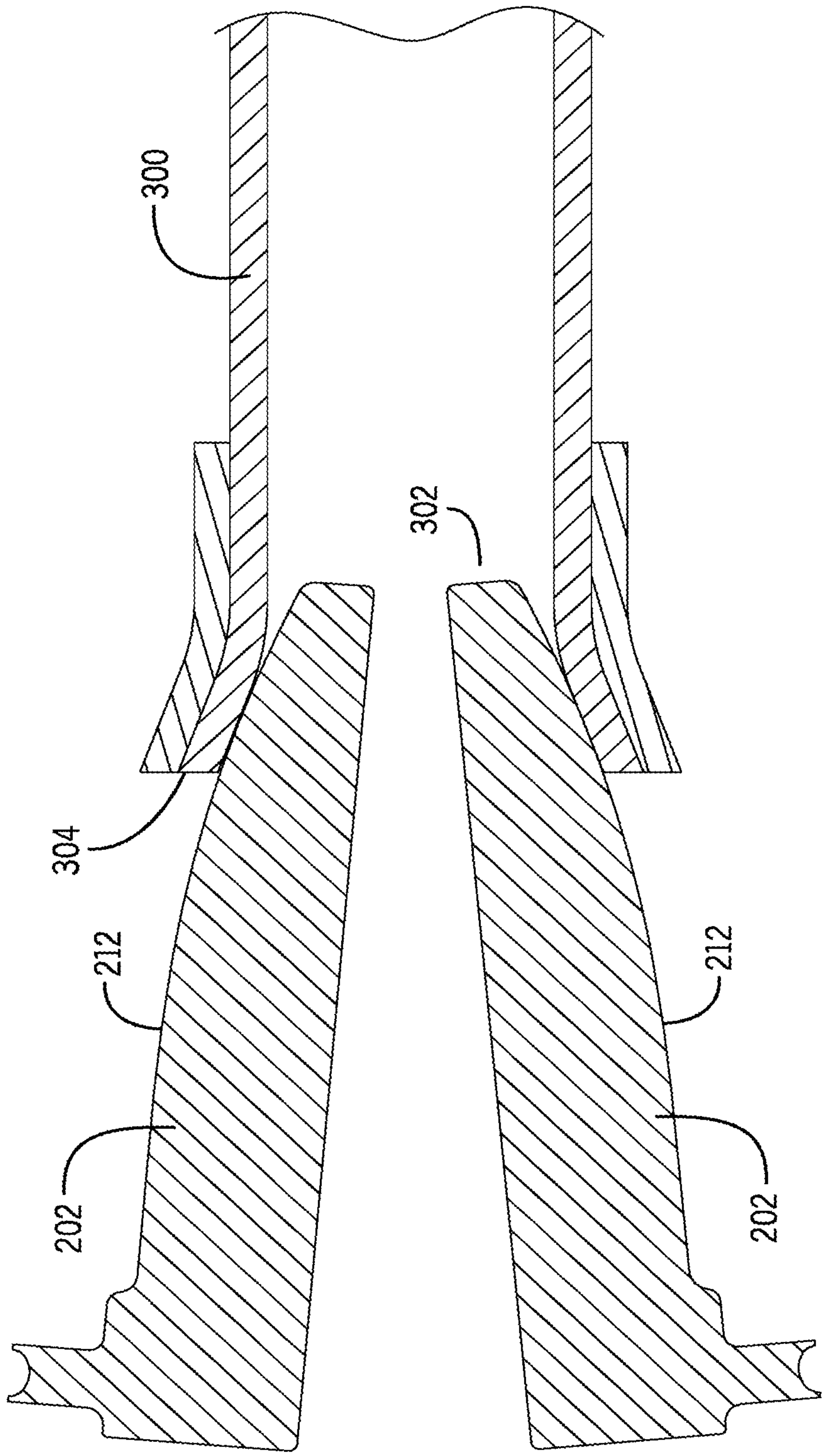


FIG. 7C

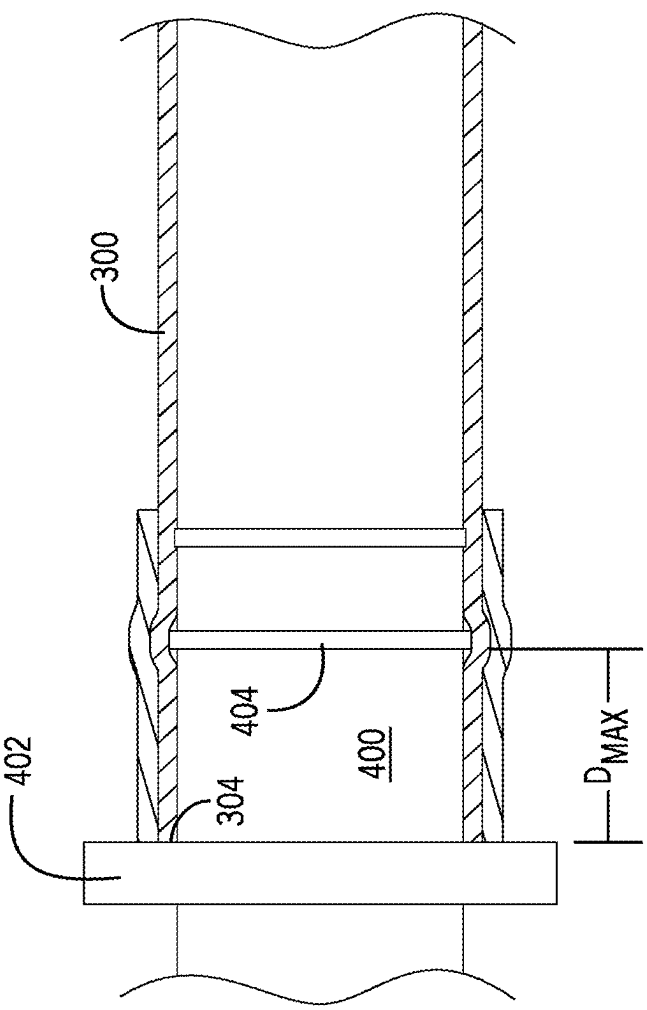


FIG. 8

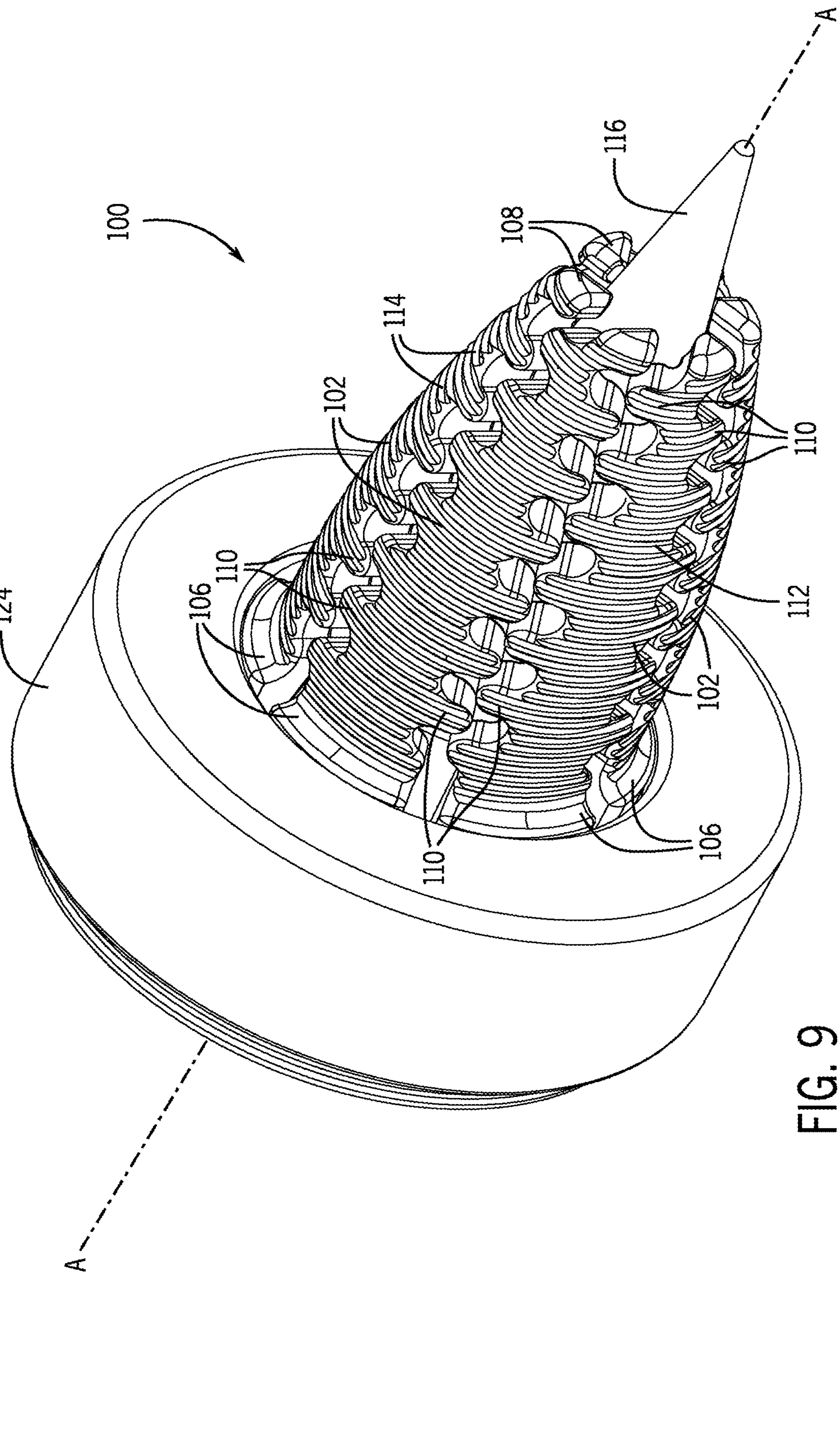


FIG. 9

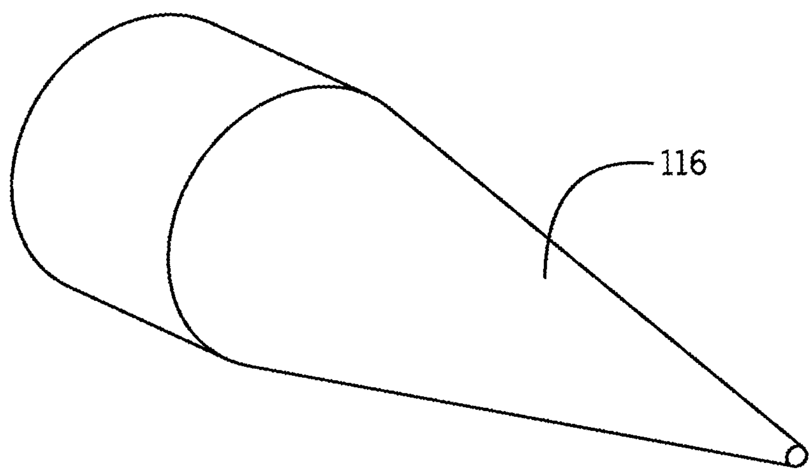


FIG. 10A

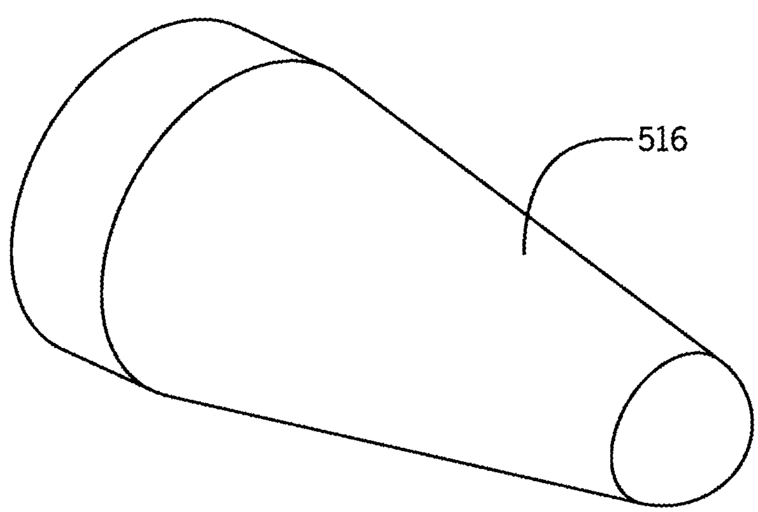


FIG. 10B

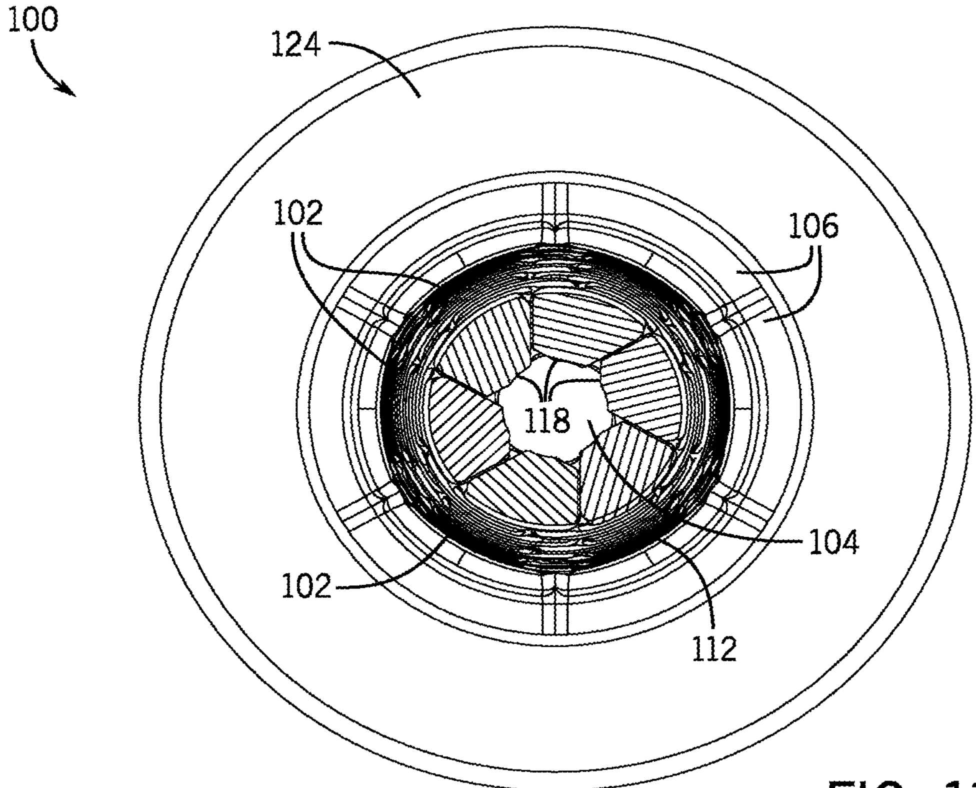


FIG. 11A

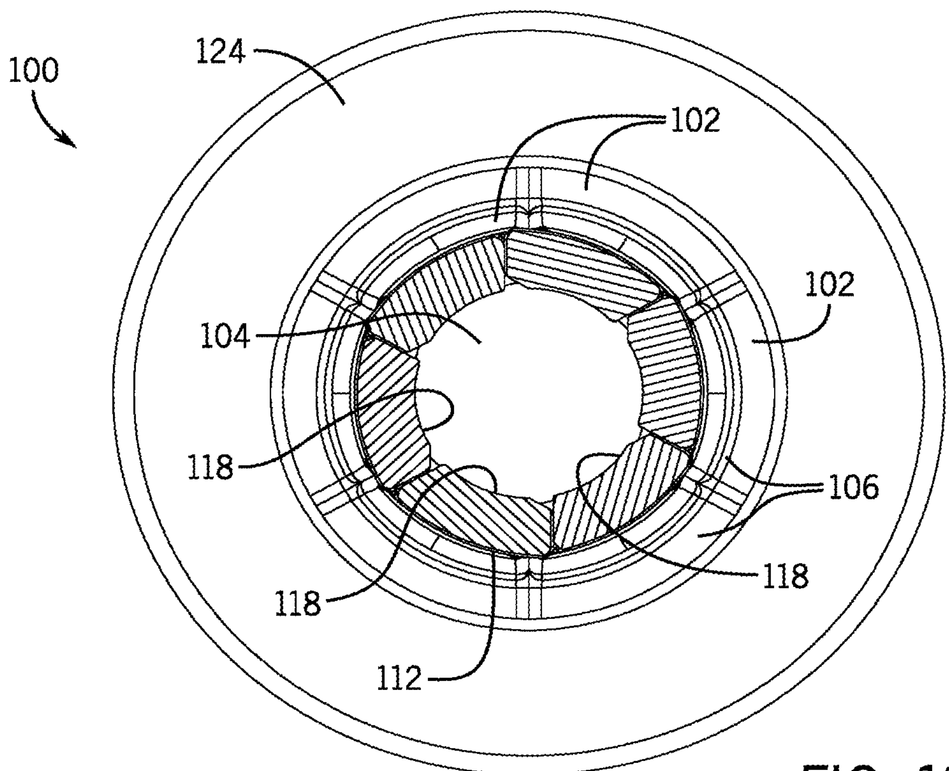


FIG. 11B

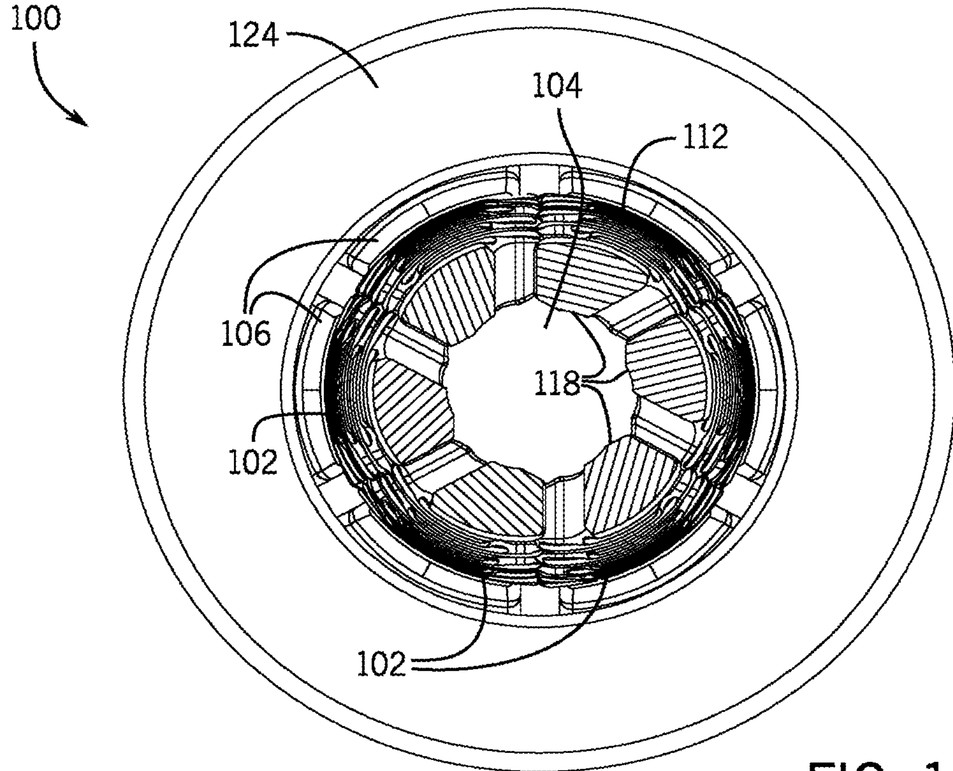


FIG. 12A

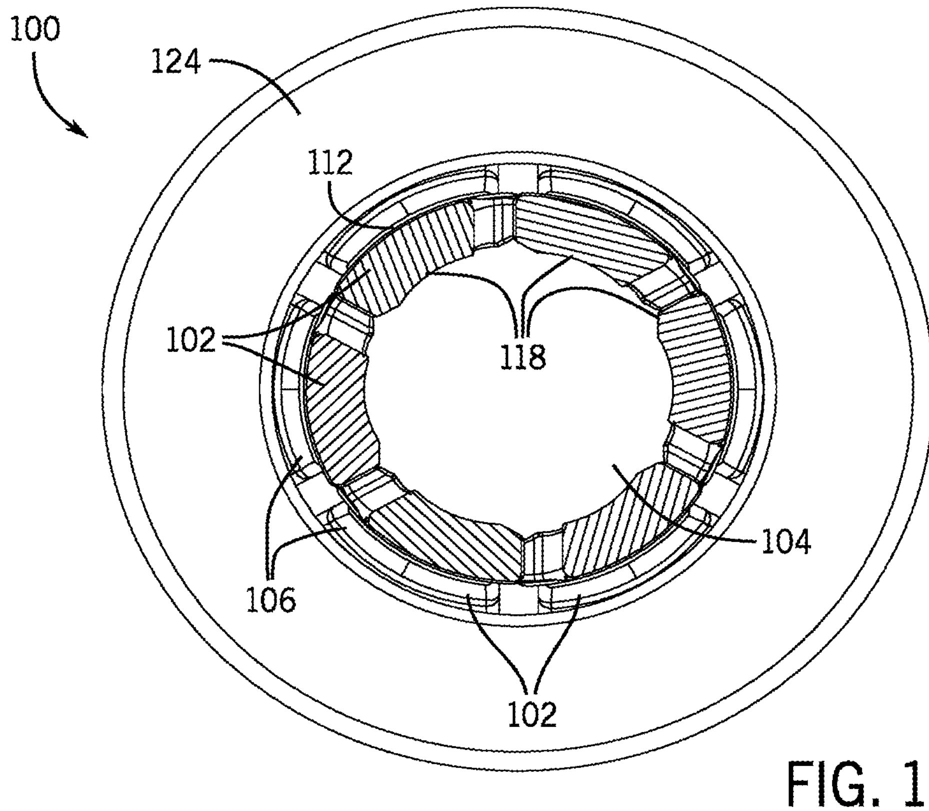
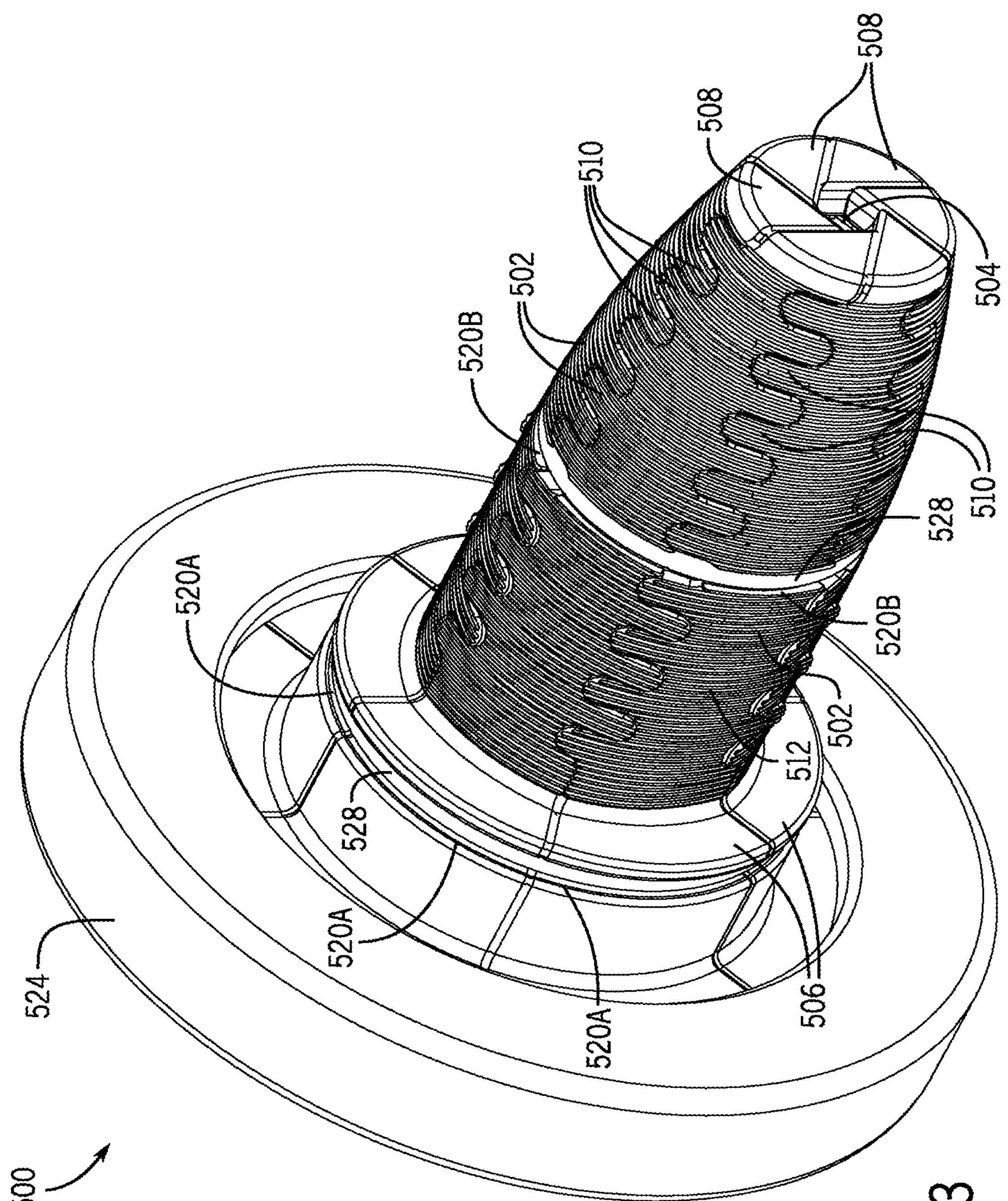


FIG. 12B


500
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520B

FIG. 13

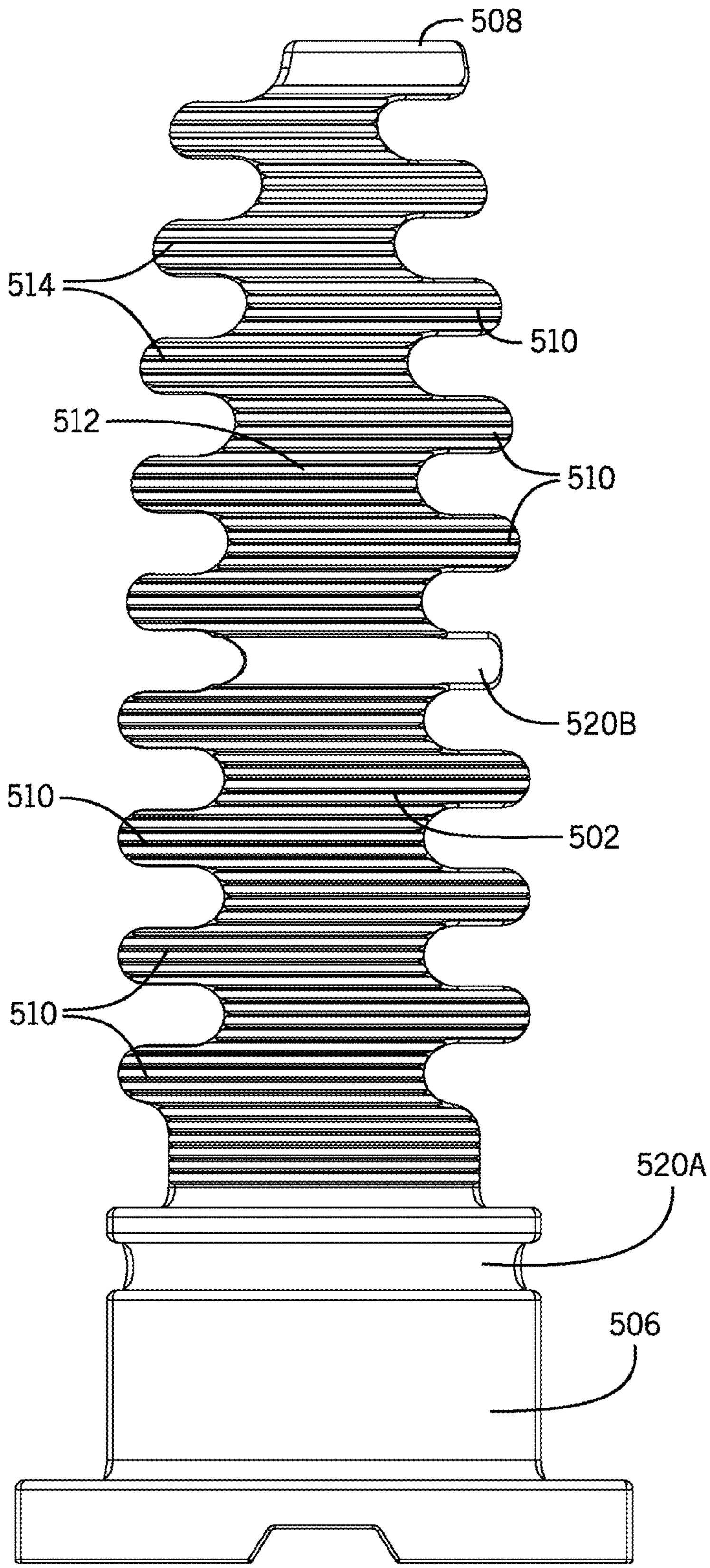

FIG. 14

CONSTANT STRAIN PEX EXPANSION TOOL HEAD

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional of co-pending U.S. patent application Ser. No. 16/393,525, filed on Apr. 24, 2019, which claims the benefit of U.S. Provisional Application No. 62/673,708 filed May 18, 2018, the entire contents of which are incorporated by reference herein.

STATEMENT OF FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not applicable.

TECHNICAL FIELD

[0003] This application relates to expansion tool heads for the expansion of PEX tubing in the assembly of PEX plumbing connections.

BACKGROUND

[0004] PEX piping or tubing is commonly used in the plumbing of residential and commercial building construction. PEX tubing is made from am extruded crosslinked polyolefin, typically high density polyethylene (HDPE). PEX pipes are coupled to each other or to fixtures at fittings designed to form a water-tight seal between a radially-inward facing side of an axial end of the PEX pipe and a radially-outward facing side of the fitting which usually includes one or more circumferential ribs. In order to connect an end of a PEX pipe to a fitting via cold expansion (typically at room temperature), the opening of the axial end of the PEX pipe—which end often further includes a collar surrounding the end—is temporarily enlarged in diameter to allow the fitting to be inserted into the opening of the PEX pipe. As the axial end of the pipe and any surrounding collar naturally return towards their un-deformed and un-expanded states, the radially-inward facing side of the PEX pipe is compressed against the radially-outward facing side of the fitting, especially the ribs, to form the water-tight seal between the pipe and fitting.

[0005] To quickly and efficiently expand the axial ends of PEX tubing in the formation of connections or couplings with fittings, expansion tools are commonly used. Expansion tools typically feature an expansion tool head with a set of expandable jaws including a frusto-conically shaped portion. In use, a portion of the jaws that are frusto-conically shaped are initially inserted into the axial end of the PEX pipe. The inserted jaws are then actuated to move the jaws outward against the inner surface of the PEX pipe so that the jaws press against the PEX pipe to stretch the walls of the pipe and temporarily increase the PEX pipe's inner diameter. This expansion of the jaws can often be repeated multiple times—by retracting the jaws to a closed position, rotating the jaws angularly relative to the tool and pipe, and then re-expanding the jaws against the inner surface of the PEX pipe—in order to sufficiently enlarge the opening of the pipe to accommodate insertion of the fitting as well as to distribute the localized strain created by the jaws around the circumference of the PEX pipe at each individual expansion step.

SUMMARY

[0006] It is herein recognized that, when these jaws are expanded to engage the inner surface of the PEX pipe, how the jaws engage the pipe to induce the strain in the pipe can alter the strain distribution at the expansion step(s) and that improvements that would better or more evenly distribute this strain might provide great advantage to the robustness of the seal formed.

[0007] Conventionally, expansion tools have jaws in which in the retracted state, the jaws form a simple frusto-conical surface as wedge-shaped sections on each of the jaws for contacting the inside diameter of the PEX pipe. However, such a basic geometry could result in uneven strain distributions during the expansion step(s).

[0008] For example, the sections of the wedges of the jaws that contact the pipe can create axial lines of contact along the jaws and axial lines of non-contact between the jaws that create axial lines of strain during expansion. The lines of axial contact can, due to friction, locally hold the material of the PEX pipe contacting the jaws such that the material close to the points of contact are not strained while the strain that is generated in the pipe is primarily and preferentially generated along the axial lines of the material where the PEX pipe is not contacted by the jaws. Although the expansion process can include multiple steps that may help to improve strain distribution by rotating the tool head relative to the pipe, it is often the case that the amount of strain induced at each expansion stroke of the tool will not be equal and the initial preferential deformation will not be equalized by subsequent expansion steps. It is noted that such rotation of the tool head relative to the pipe may be manual (i.e., the user may rotate a manually-actuated tool or the pipe to attempt to equalize the induced circumferential strain which occurs during expansion) or automatic (i.e., a power tool may automatically rotate the tool head some amount between steps). However, if such rotation is not made or is attempted but improperly executed, the axial linear strain of adjacent expansion steps can be exacerbated or stacked.

[0009] Still further, if the wedges are frusto-conically shaped, then a distal tip of the jaws may dig into the radially-inward facing surface of the pipe (especially on the first expansion stroke) creating some amount of preferential deformation including some amount of deformation that may be plastic or permanent in nature rather than elastic. If this plastic deformation occurs at a depth in the opening of the PEX pipe corresponding to the sealing ribs of the fitting that the pipe is received upon, then that or those regions of plastic deformation may provide potential leak pathways from the inner volume of the pipe between the fitting and the pipe where portions of the pipe have been excessively plastically deformed—especially when the deformation creates a pathway axially past the sealing rib of the fitting.

[0010] Various improved designs for expansion tool heads are disclosed herein that are structured (1) to provide improved and more-evenly distributed strain in the PEX pipe during the expansion strokes of the tool head and/or (2) to avoid the induction of plastic deformation of the PEX pipe in regions of seal formation between the radial-inwardly facing surface of the PEX pipe and the radially-outward facing surface of the fittings (i.e., the ribs of the fitting). Although not limited to merely these aspects, the improved designs can incorporate replacing the traditional frusto-conical surface of the jaws with a continuously curved convex surface in the axial direction such that a sharp-edge of the frusto-conical shape does not dig into the inner wall of the pipe. Instead, a softer convex intermediate part of the jaw profile may initially engage the inner diameter of the pipe during expansion. Still yet, the profile of the jaws may be structured to limit the initial axial insertion depth of the jaws into the opening to prevent permanent deformation of the PEX pipe in the region where the seal with the rib of the fitting will be formed. The jaws might alternatively or additional have angularly-extending interdigitating teeth that help to inhibit the formation of axially-extending strain lines. While such expansion tool heads have been known to include interdigitating teeth, herein the structure of the teeth may be altered in a novel manner such that, near the tip of the jaws, the teeth form a true round profile in the retracted or closed position of the jaws and such that, near the base of the jaws, the teeth form a true round profile in the extended or opened position of the jaws. This hybrid profile in which the profile of the teeth vary over the axial distance can be advantageous to avoid the possibility that the teeth at the tip end dig into the walls of the PEX pipe during the initial expansion step (as might be the case if they were true round in the fully opened position of the jaws).

[0011] Still further, these new expansion tool head geometries may stretch the pipe less at the region outside of the area covered by an expansion ring, thereby allowing for a fitting to be inserted with less potential damage/strain on the pipe adjacent to the fitting. This can also have the benefit of reducing the time it takes for the fitting to shrink back on the pipe.

[0012] Still yet another improvement may be that, because of the continuously curved convex geometry or shape, it may be more difficult to force the tool into the pipe (effectively, avoiding an initial over-insertion condition of the tool head in the pipe which, upon expansion could result in immediate and drastic over-expansion of the pipe at a rate and amount exceeding what the first expansion should entail). Instead, the tool head can be ensured to more gradually perform the work appropriate to induce the strain on the pipe, rather than having the potential for over-insertion and over-expansion, especially at a first expansion step, which may damage the pipe.

[0013] According to one aspect, the expansion tool head includes a plurality of jaws arranged circumferentially around a central axis. Each of the jaws extends from a base end to a tip end and has a set of angularly-extending teeth. The teeth of each of the jaws interdigitate with the teeth of the circumferentially adjacent jaws. The jaws are movable between a retracted position in which the jaws are brought together with one another and an expanded position in which the jaws are separated from one another. The jaws collectively provide a curved outer surface that has a profile which, notably, varies over an axial length of the jaws such that, when the jaws are in the expanded position, the curved outer surface of adjacent jaws are tangent to one another at a first axial position proximate to the base end and, when the jaws are in the retracted position, the curved outer surface of adjacent jaws are tangent to one another at a second axial position proximate to the tip end. Among other things, this can help to avoid axial stain from developing and further helps to avoid the teeth from plastically digging into the pipe at the tip end during the initial expansion stroke or strokes of the jaws.

[0014] According to another aspect, an expansion tool head includes a plurality of jaws extending from a base end to a tip end in which the jaws are again arranged circumferentially around a central axis. The jaws are movable between a retracted position in which the jaws are brought together with one another and an expanded position in which the jaws are separated from one another. According to this aspect, a curved outer surface collectively provided by the jaws tapers radially inward in the axial direction from the base end to the tip end in a continuous curve which is convex in form (at least in part) relative to the central axis of the jaws. With such a convex continuous curved profile of the jaws, an edge at the axial end of the jaw can be inhibited from digging into the material of the pipe, especially upon the first expansion stroke.

[0015] According to yet another aspect, a method for expanding an axial end of a pipe to accommodate reception of a fitting is disclosed in which the fitting has an abutment flange for positioning the axial end of the pipe on the fitting and a rib for making a sealed connection in which the rib that is separated from the abutment flange by a pre-established distance. A tip end of an expansion tool head is inserted into the axial end of the pipe. Notably, the expansion tool head includes a set of jaws that are collectively shaped with respect to an opening of the pipe in order to limit an insertion distance of the tip end of the expansion tool head into the axial end of the pipe to an amount less than the pre-established distance between the abutment flange and rib. The jaws are then actuated to move the jaws outward from a retracted position (in which the jaws are positioned together with one another) to an expanded position (in which the jaws are separated from one another) thereby expanding the axial end of the pipe. By limiting the insertion depth of the jaws on the first stroke, it may be ensured that permanent plastic deformation of the PEX pipe does not occur in the region of the rib of the fitting.

[0016] It is contemplated that the various improvements to the geometric shape of the jaws may enable the tool have to have jaw actuation patterns different than jaw opening patterns which are equally radial over the axial length. Instead, with the geometries described herein, it is contemplated that, over the actuation used to expand to the jaws, the tip end of the tool may remain together while the base end of the jaws are separate from one another. The tip end could remain together entirely, or simply be actuated outward to a lesser extent than the base end. This style of expansion (i.e., in which the separation of the base end of the tool exceeds that at the tip end) can mean that the expansion of the pipe can be more localized to the axial length over which the fitting will be received (i.e., the axial end alone without extending deeper into the tube) and can mean that expansion tool head may achieve the objective of the expansion with a reduced number of expansions.

[0017] These and still other advantages of the invention will be apparent from the detailed description and drawings. What follows is merely a description of some preferred embodiments of the present invention. To assess the full scope of the invention, the claims should be looked to as these preferred embodiments are not intended to be the only embodiments within the scope of the claims.

BRIEF DESCRIPTION OF THE FIGURES

[0018] FIG. **1** is perspective view of an expansion tool head having a hybrid profile with interdigitated teeth in a closed or retracted position.

[0019] FIGS. **2**A and **2**B are side views of a single jaw of the expansion tool head of FIG. **1**.

[0020] FIGS. **3**A and **3**B are side views of a single jaw of an expansion tool head different than that of the expansion tool head of FIG. **1** having no teeth, but a continuous curved convex shape type profile over its axial length.

[0021] FIG. **4** is a top-down view along the central axis of the expansion tool head of FIG. **1** in a retracted or closed position.

[0022] FIG. **5** is a top-down view along the central axis of the expansion tool head of FIG. **1** in an expanded or opened position.

[0023] FIG. **6** is a cross-sectional side view of an expansion tool head incorporating the jaws of FIGS. **3**A and **3**B in which the expansion tool head is in the retracted position in which it is initially positioned within a PEX pipe before expansion.

[0024] FIGS. **7**A-**7**C are cross-sectional views of the expansion tool head within the PEX pipe as in FIG. **6**, but in which the jaws have been moved to an expanded position within the PEX pipe to expand it. FIG. **7**A shows an expansion in which the jaws are moved outward in a nearly purely radial translational fashion, while FIG. **7**B shows a variation in which the tip ends remain substantially together while the base ends are separated, and FIG. **7**C shows another variation in which both the base ends and the tip ends of the jaws are separated radially from one another but the extent of separation of the tip ends from one another are less than the base ends.

[0025] FIG. **8** is a cross-sectional view of a PEX pipe having been expanded from FIGS. **6** and **7** after return towards its initial size to form a water-tight seal with a fitting.

[0026] FIG. **9** is a perspective view of the expansion tool head of FIG. **1** in an open or expanded position.

[0027] FIGS. **10**A and **10**B are perspective views of jaw actuators for moving an expansion tool head between a retracted position and an expanded position. FIG. **10**A shows a jaw actuator with a conical profile and a narrow rounded tip. FIG. **10**B shows a jaw actuator with a conical profile and a wide rounded tip.

[0028] FIGS. **11**A and **11**B are cross-sectional front views of an expansion tool head without the jaw actuator in which the expansion tool head is in the retracted position. FIG. **11**A is a cross section taken near the tip end of the jaws. FIG. **11**B is a cross section taken near the base end of the jaws.

[0029] FIGS. **12**A and **12**B are cross-sectional views of the expansion tool head of FIGS. **11**A and **11**B, but in which the jaws have been moved to an expanded position.

[0030] FIG. **13** is perspective view of an expansion tool head having a hybrid profile with interdigitated teeth in a retracted position and a positioning band between the tip and the base of the jaws.

[0031] FIG. **14** is a side view of a single jaw of the expansion tool head of FIG. **13**.

DETAILED DESCRIPTION

[0032] Embodiments of the disclosure may be further understood with reference to the figures.

[0033] FIG. **1** illustrates an exemplary embodiment of an expansion tool head **100**. The expansion tool head **100** includes a plurality of jaws **102** arranged circumferentially around a central opening **104** and a central axis A-A. The jaws **102** each extend from a base end **106** to a tip end **108**, and are movable between at least two positions-a retracted position as shown in FIGS. **1** and **4** and an expanded position as shown in FIGS. **5** and **9**. As illustrated, the expansion tool head **100** includes six identical jaws **102**; however, in other expansion tool heads, there may be other multiples of jaws.

[0034] Illustrated in additional detail in FIGS. **2**A and **2**B, each of the jaws **102** include a plurality of angularly-extending teeth **110** disposed along an axial length of each jaw **102** between the base end **106** and the tip end **108**. The teeth **110** are arranged in a repeating, alternating pattern in which the teeth **110** alternately extend from the jaw **102** in a first angular direction then extend from the jaws in a second, opposite angular direction, and so forth as one travels down the axial length of the jaw **102**. The tooth **110** placement pattern is substantially identical on each of the jaws **102** such that the teeth **110** on each jaw **102** interdigitate with the teeth **110** of the circumferentially-adjacent jaws **102**. In other non-limiting embodiments, the teeth **110** can be arranged in alternative patterns or directions or the jaws may be differently-shaped, but still complimentary with one another.

[0035] Referring back to FIG. **1**, the jaws **102** collectively provide a curved outer surface **112** which faces outward and away from the central axis A-A. In some embodiments, a profile of the curved outer surface **112** varies over at least a portion of the axial length of the jaws **102**, while in other embodiments a profile of the curved outer surface may remain constant over a portion of the axial length of the jaw. As illustrated in all of the illustrated jaws, the curved outer surface tapes radially inward in the axial direction (from base end **106** to tip end **108**) so that the curved outer surface on any particular jaw forms a continuous curve which is generally convex in shape (save for any de minimus circumferential ridges **114**). Because any ridges **114** are circumferential, in the depicted embodiments, they necessarily have sections on all of the jaws **102** (in this case, all six of the jaws) at the same axial position. In other embodiments, however, it is contemplated that circumferential ridges may be omitted from the curved outer surface.

[0036] To move the expander tool head **100** between a retracted and an expanded position, the tool upon which the expander tool head **100** is attached further includes a jaw actuator **116** (as shown in FIG. **10**A. In the embodiment illustrated, the jaw actuator **116** is a conical spindle positioned concentrically with the central axis A-A and the central opening **104**. The conical spindle or jaw actuator **116** is axially translated into the central opening **104** and into contact with a cooperating inner surface **118** of the jaws **102**. The conical spindle or jaw actuator **116** pushes the jaws **102**, at least in part, radially outward and away from the central axis A-A into an expanded position, as shown in FIG. **5**. In the expanded position, each of the jaws **102** are angularly separated from circumferentially-adjacent jaws **102**. In some embodiments, an alternative jaw actuator can be used. For example, FIG. **10**B shows a jaw actuator **516** with a blunt tip that is wider and rounded with a larger radius than the tip of the jaw actuator **116** of FIG. **10**A. Some embodiments can include other jaw actuators that are differently sized or differently shaped than the illustrated jaw actuators.

[0037] To help delimit the range of motion during expansion of the jaws **102** and with further reference being made to FIGS. **1-2**B, the base end **106** of the each of the jaws **102** includes a groove **120** and can be surrounded by a retention bracket or restrictive collar **124** which is attached to the tool.

[0038] The groove **120** which is arranged perpendicular to the central axis A-A and centered thereabout and is configured to receive a positioning ring or band (see, e.g., the band **528** in another embodiment illustrated in FIG. **13**) to ensure that the jaws **102** are connected or bundled to one another at the base end **106** to prevent or inhibit separation at that location. The positioning ring can be clastic and biases the jaws **102** into a retracted position shown in FIGS. **1** and **4**. In the retracted position, each of the jaws **102** are drawn toward the central axis A-A and one another, such that in the illustrated form of FIGS. **1**, **2**A, **2**B, **4**, **5**, and **9**, the teeth **110** of the jaws **102** are positioned together so that there are substantially no gaps in-between the jaws **102**. In this position, the circumferential ridges **118** proximate the tip end **108** form a continuous ridge around the curved outer surface **114**.

[0039] To hold this bundled group of jaws **102** on the greater tool, a retention bracket or restrictive collar **124** can be positioned proximate the base end **106**. This collar **124** can both secure the jaws **102** on the tool and can be further configured to restrict axial or tilting movement of the jaws **102** during their expansion.

[0040] As the jaws **102** move from the retracted position to the expanded position by engagement with the jaw actuator or conical spindle **116** with the cooperating inner surfaces **118**, each of the jaws **102** are moved outward relative to the central axis A-A such that the jaws **102** and the tip end **108** of each of the jaws **110** most dramatically are moved radially outward and away from each other. This will be used to expand the inner surface PEX tubing. Then, when the jaw actuator or conical spindle **116** is retracted, the positioning ring in the groove **120** biases the jaws **102** inward to the retracted position drawing the jaws **102** together again.

[0041] During the expansion of PEX tubing, this expansion and retraction can happen in various times in progression. Between each expansion stroke, the tool can be designed such that the jaws **102** can rotate between each expansion stroke. In this way, the strain induced by each expansion step can be distributed over different parts of the volume of the PEX material.

[0042] Having described the general shape of the jaws **102** from FIGS. **1**, **2**A, **2**B, **4**, **5**, and **9**, some of the advantages of the particular jaw geometry can now be described and appreciated. Most notably, the continuously curved convex surface from one axial end to the other can help to inhibit permanent deformation in the PEX material and a varying profile of the teeth **110** can help to distribute strain in non-linear fashion along axial lengths also without inducing undesirable plastic deformation.

[0043] First, given the convex, continuously curved surface from a region proximate the base end **106** to a region proximate the tip end **108** and over the axial length of the PEX-contacting area of the jaws **102**, when the jaws **102** are opened or expanded, they contact the inside of the PEX tube in a manner such that a sharp edge is not dug into the wall of the tube. In some forms, this may mean that a rounded ring between the base end **106** and the tip end **108** first contacts the inside of the pipe in a non-destructive fashion. This may mean, for example, that in the expanded position, a radially outermost ring of the curved outer surface **112** is between the base end **106** and the tip end **108** and is not at a sharp edge or discontinuity that could destructively dig into the PEX material as would be the case in the standard frusto-conical jaw design. However, in other forms, a ring of contact between the tip end **108** of the jaws **102** and the inside of the PEX tube may occurs first, but that the convex profile will subsequently even out the localized deformation of the PEX material at the tip end **108**.

[0044] With respect to the profile of jaws **102** over their axial length, the curved outer surface **112** of the jaws **102** and their respective teeth **110** differs or varies over the axial length. At a first axial position proximate the base end **106**, the curved outer surface **112** of adjacent jaws **102** are tangent with one another when the jaws **102** are in the expanded position. However, as can be seen in FIG. **4** in the retracted position, the ends of those teeth **110** protrude radially outward. At a second axial position proximate the tip end **108**, the curved outer surface **112** of adjacent jaws **102** are tangent only when the jaws **102** are in the retracted position. This structure can advantageously provide that, in the expanded position, especially near the base end **106**, the inside of the PEX pile can have a large amount of surface area contact with the jaws **102**, including up to the angular edges of the teeth **110**. However, as can be seen from FIG. **4**, to achieve such a circular profile when expanded, in a less than fully expanded state, these teeth **110** can protrude and potentially could provide elements which cause undesirable deformation in the PEX material. To avoid this on the tip end **108** (which will engage the PEX while it is still in an initially un-deformed state), the profile at the tip end can be circular in the retracted position, thereby preventing the possibility that the edges of the teeth piece or deform the PEX material.

[0045] FIGS. **3**A and **3**B illustrate an alternative jaw **202** which might be used in place of jaw **102** in an alternative expansion tool head. The alternative jaw **202** is similar to jaws **102** in that it has a continuous curved surface from the base end **206** to the tip end **208** which is convex relative to the central axis. However, it does not include any teeth. As with the first embodiment, this second embodiment can help to avoid digging into the PEX material and its shape can prevent or limit insertion of the tool head into a pipe as will be described below. It is noted that all features on FIGS. **3**A and **3**B are labeled using the **200** series numbers and like numerals from the **100** series are used to denote like features described herein. For example, the jaw **202** includes a groove **220**, an inner surface **218**, and circumferential ridges **214**.

[0046] Additionally, FIGS. **13** and **14** illustrate an alternative expansion tool head **500** with a set of alternative jaws **502** that may be used in place of expansion tool head **100**. The alternative expansion tool head **500** is similar to expansion tool head **100** in that it has a continuous curved surface from the base end **506** to the tip end **508** which is convex relative to a central axis. However, each jaw **502** includes a first groove **520***a* positioned proximate the base end **506** and a second grove **520***b* positioned between the base end **506** and the tip end **508**. Each of the grooves **520***a*, **520***b* is configured to receive a positioning ring **528** for biasing the jaws **502** into the retracted position. The use of additional positioning rings can help to bias differently sized or shaped jaws into a retracted position, and can help limit the movement of the jaws as the move into an expanded position. It is noted that all features on FIGS. **13** and **14** are labeled using the **500** series numbers and like numerals from the **100** series are used to denote like features described herein. For example, the jaw **502** includes a collar **524**, a curved outer surface **512**, teeth **510**, and circumferential ridges **514**.

[0047] With reference to FIGS. **4**, **5**, and **11**A-**12**B it can be seen that the profile of the curved outer surface **112** of the jaws **102** can be shaped such that the profile is true round in sections in the closed position—as can be seen by the true circle at the tip end in FIGS. **4** and **11**A—and/or true round (save for any gaps between the teeth) in the opened position—as can be seen by the true circle at the base end in FIGS. **5** and **12**B.

[0048] As illustrated this "true round" condition includes the entirety of the outer surface of the jaws including the radially-outward facing surface of any teeth. Because the tool head of FIGS. **4**, **5**, and **11**A-**12**B is a hybrid design in which the cross-sectional profile varies over the axial length, it includes both true round configurations in different axial sections in both the opened and closed position of the jaws. It is noted that, when true round is achieved in the extended position and there are interdigitating teeth, these teeth may protrude or stick out from the outer surface in the retracted position as illustrated with the base end teeth in FIGS. **4** and **11**B. Although seemingly harsh-looking, because these protrusions disappear as the tool head opens, they are harmless to the inside of the pipe.

[0049] Numerous variations are contemplated as to when alternative forms of jaws can be true round. For example, all or some of the outer surface could be true round either in the retracted or the extended position. Of course, when a section is true round in one of those positions, it will not be in the other. Still further, as the actuation modality of the jaws can varied (as is illustrated between FIG. **6** and the alternative extended positions in FIGS. **7**A-**7**C), the outer surface profile may be varied such that true round is only achieved in the final opened expanded position on some or all portions of the surface of the jaws based on the desired orientation and configuration of the jaws. Still further, the true round condition could be present on jaw designs have convex curved surfaces over their axial lengths or on jaws having merely straight profiles in either the opened or closed position of the jaws.

[0050] FIGS. **6** and **7**A-**7**C schematically illustrate various potential operating positions of the expansion tool head in use to expand a PEX pipe **300** using an expansion tool head employing jaws like jaws **200** above illustrated in FIGS. **3**A and **3**B.

[0051] In FIG. **6**, the jaws **202** of the expansion tool head have been inserted into an opening **302** of an axial end **304** of the PEX pipe **300** and the jaws **202** are in the retracted position. As illustrated, axially-facing surface of the axial end **304** is in contact with the curved outer surface **212** at a contact ring on the continuous curve. The particular set of jaws **202** and expansion tool head should be selected such that the initial insertion distance DI (i.e., the distance from the axial end **304** to the tip end **204** of the jaws **202**) is less than the pre-established distance DMAX (e.g., as illustrated in FIG. **8**) from the abutment flange **402** of the fitting **400** to the rib **404** of the fitting. This limit on insertion depth precludes any part of the jaws **202** from plastically deforming the inside of the PEX pipe **300** in a region that will be compressed around the rib **404**.

[0052] Looking at FIGS. **7**A-**7**C, the jaws **202** are shown in an expanded state in which the jaws **202** have been used to enlarge the opening **302** proximate the axial end **304** of the PEX pipe **300**. As noted above, this expansion may happen in various iterative steps. Once the PEX tube **300** is expanded sufficiently, then the PEX tube **300** can be slid over the fitting **400** such that axial end **304** of the PEX tube **300** abuts an opposing axial face of the abutment flange **402** of the fitting **400**. The expanded PEX material of the PEX tube **300** is then allowed to naturally return towards its original shape and dimensions over a duration of time (often on the order of magnitude of 10 seconds to a few minutes) to form a water-tight seal around the fitting **400**.

[0053] Notably, the jaws **202** might be actuated from the inserted, retracted position of FIG. **6** to any of the expanded positions of FIG. **7**A, **7**B, or **7**C. For the sake of clarity, the positions shown in FIGS. **7**A, **7**B, and **7**C are not sequential, but rather are alternative final expanded positions for the jaws.

[0054] It can be seen that the primary difference between the positions illustrated in FIGS. **7**A, **7**B, and **7**C are the manner in which the jaws **202** are expanded from one another. In the actuated position shown in FIG. **7**A, the jaws **202** are simply translated radially outward such that the base end and the tip end of each of the jaws are equally translated away from the central axis. In this type of movement, the jaws in FIG. **7**A are not rotated with respect to the central axis. In the actuated position shown in FIG. **7**B, the jaws **202** are rotated such that the base ends are separated from the central axis while the tips of the jaws **202** remain together at the central axis. In the actuated position shown in FIG. **7**C, the jaws **202** are rotated so that both the base end and the tip end are separated from the central axis, with the base end being separated to a greater extent than the tip end.

[0055] All of these actuation modalities of the jaws **202** can help to expand the tube **300** and compression collar in a more beneficial way for inserting the fitting **400** as will be depicted in FIG. **8**. Notably, the curved profile and actuation modalities can help focus the deformation so that it that primarily occurs at the axial end (where the fitting will be inserted) rather than in a region past the compression collar which is less well-supported. Further, especially in the expansion positions illustrated in FIGS. **7**B and **7**C, the "tip in" configurations help to reduce the rate of expansion that the jaws **202** perform on the end of the pipe.

[0056] Referring now to FIG. **8**, this connection between the fitting **400** and the recoiled PEX tube **300** is illustrated. Notably, the rib **404** protrudes outward from the fitting **400** to cause the primary seal and given the earlier stated limitation on tool insertion distance during the insertion and expansion stage, it can be ensured that no irreversible plastic deformation has occurred in this region.

[0057] While the expansion of PEX tubing using expansion tool heads having various geometric profiles have been described, it is contemplated that tubular materials other than PEX tubing could also be expanded using these expansion tool heads. For example, copper or other metal tubes may also be able to have axial ends thereof expanded using tool head geometries of this type.

[0058] While various representative embodiments of improved expansion tool head geometries have been illustrated, that many general principles are contemplated as being independently employable as well as in all workable permutations and combinations. For one, the improvement of a continuous curved convex shape is considered novel, especially as a means for limiting insertion depth into a pipe and enabling alternative actuation modes for the jaws. Such curved geometry could be widely employable in jaw sets having straight sides, sides with interdigitated teeth, or even other patterns (for example, spiral or helical jaws). Still further, the designs with interdigitating teeth could have a curved profile in the axial direction or a straight profile over their axial length.

[0059] For both designs incorporating interdigitating teeth and designs not incorporating interdigitating teeth, the working end of the jaws could have a profile in a cross-section taken perpendicular to the central axis that is circular when the jaws are closed or circular when the jaws are opened. A particular jaw could have such profiles (circular when opened or circular when closed) over all or just a part of the axial length and it is further contemplated that an expansion tool head can have a hybrid form in which it shifts from one type to the other type over the axial length of the tool head as depicted in FIGS. **1** and **13**, for example.

[0060] Still further serrations or ridges may be present entirely over the working end of the jaws or over just a portion thereof (i.e., just proximate the tip). Again, such serration could be present on any of the various permutations from the variations apparent from the paragraphs above.

[0061] Still yet, it is contemplated that any of the three actuation modalities illustrated above in FIGS. **7A-7C** could be used with these different types of jaw designs and permutations. It will undoubtedly be appreciated that certain of these actuation modalities can present advantages over others based on the usage context in which the expansion tool head is used and that, given a particular pipe diameter and actuation modality, the jaw might be engineered to include various ones of the these design aspects in a particularly advantageous manner. Thus, while this disclosure provides many representative jaw structures, the structures contemplated within this disclosure are not so limited.

[0062] It should be appreciated that various other modifications and variations to the preferred embodiments can be made within the spirit and scope of the invention. Therefore, the invention should not be limited to the described embodiments. To ascertain the full scope of the invention, the following claims should be referenced.

What is claimed is:

1. An expansion tool head for an expansion tool, the expansion tool head comprising:

a plurality of jaws extending from a base end to a tip end in which the jaws are arranged circumferentially around a central axis and each have a set of teeth interdigitating with the teeth of circumferentially adjacent jaws, the jaws providing a curved outer surface, and the jaws being movable between a retracted position in which the jaws are brought together with one another and an expanded position in which the jaws are separated from one another; and

wherein a profile of the curved outer surface of the jaws varies over an axial length of the jaws such that, when the jaws are in the expanded position, the curved outer surface of adjacent jaws are tangent to one another at a first axial position proximate to the base end, and such that, when the jaws are in the retracted position, the curved outer surface of adjacent jaws are tangent to one another at a second axial position proximate to the tip end.

2. The expansion tool head of claim **1**, wherein the jaws further include circumferential ridges on the curved outer surface.

3. The expansion tool head of claim **2**, wherein the circumferential ridges are positioned on a first portion of the curved outer surface and a remainder of the curved outer surface is smooth.

4. The expansion tool head of claim **3**, wherein the circumferential ridges on the first portion of the curved outer surface are located proximate the tip end and the curved surface proximate the base end is smooth.

5. The expansion tool head of claim **1**, wherein the curved outer surface of the jaws tapers radially inward in an axial direction from the base end to the tip end in a continuous convex curve.

6. The expansion tool head of claim **1**, wherein, when the jaws are in the retracted position, a first profile of the curved surface is larger than a first diameter of a pipe at a first axial position proximate to the base end, and a second profile of the curved surface is smaller than a second diameter of the pipe at a second axial position proximate to the tip end.

7. The expansion tool head of claim **1**, wherein, in the retracted position, each of the jaws are drawn toward the central axis, and, in the expanded position, each of the jaws are moved radially outward and away from the central axis.

8. An expansion tool head for an expansion tool, the expansion tool head comprising:

a plurality of jaws extending from a base end to a tip end in which the jaws are arranged circumferentially around a central axis, the jaws providing a curved outer surface having a continuously curved outer profile that protrudes in a convex manner outwardly from the central axis, and the jaws being movable between a retracted position in which the jaws are brought together with one another and an expanded position in which the jaws are separated from one another, the outer surfaces of the jaws maintaining a rounded cross-section relative to the central axis in both the retracted position and the expanded position; and

wherein a profile of the curved outer surface of the jaws varies over an axial length of the jaws such that, when the jaws are in the expanded position, the curved outer surface of adjacent jaws are tangent to one another at a first axial position proximate to the base end and such that, when the jaws are in the retracted position, the curved outer surface of adjacent jaws are tangent to one another at a second axial position proximate to the tip end.

9. The expansion tool head of claim **8**, wherein the jaws further include circumferential ridges on the curved outer surface.

10. The expansion tool head of claim **9**, wherein the circumferential ridges are positioned on a first portion of the curved outer surface and a remainder of the curved outer surface is smooth.

11. The expansion tool head of claim **10**, wherein the circumferential ridges on the first portion of the curved outer surface are located proximate the tip end and the curved outer surface proximate the base end is smooth.

12. The expansion tool head of claim **8**, wherein the outer profile has a parabolic shape extending in an axial direction along the central axis, and wherein the outer profile is at least partially formed by the curved outer surface of the jaws tapering radially inward in the axial direction from the base end to the tip end in a continuous convex curve.

13. The expansion tool head of claim **8**, wherein, when the jaws are in the retracted position, a first profile of the curved surface is larger than a first diameter of a pipe at a first axial position proximate to the base end and a second profile of the curved surface is smaller than a second diameter of the pipe at a second axial position proximate to the tip end.

14. The expansion tool head of claim **8**, wherein, in both the retracted position and the expanded position, the outer surfaces of the jaws maintain a true-round cross-section relative to the central axis.

15. An expansion tool head, the expansion tool head comprising:

a plurality of jaws extending from a base end to a tip end in which the jaws are arranged circumferentially around a central axis, the jaws providing a curved outer surface, and the jaws being movable between a retracted position in which the jaws are brought together with one another and an expanded position in which the jaws are separated from one another; and

wherein the curved outer surface of the jaws tapers radially inward in an axial direction from the base end to the tip end in a continuous convex curve, the curved outer surface of the jaws having an outer profile that protrudes in a convex manner outwardly from the central axis.

16. The expansion tool head of claim **15**, wherein at least a portion of the jaws are configured to move radially outward such that, in the retracted position, the tip end of the jaws are together and such that, in the expanded position, the tip end of the jaws are separated and a radially outermost point of the curved outer surface is on the continuous convex curve between the base end and the tip end.

17. The expansion tool head of claim **15**, wherein the jaws further include circumferential ridges on the curved outer surface.

18. The expansion tool head of claim **15**, wherein, in both the retracted position and the expanded position, the outer surfaces of the jaws maintain a true round cross-section relative to the central axis.

19. The expansion tool head of claim **15**, wherein the outer profile has a parabolic shape relative to the central axis.

20. The expansion tool head of claim **15**, wherein the jaws are designed to move the base end of each jaw a first radial distance relative to the central axis and move the distal end of each jaw a second radial distance relative to the central axis, and wherein the second radial distance is less than the first radial distance.

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